firms. Both types are risk neutral. The size of the labour force is normalized to unity, whereas the number of firms will be determined endogenously by free entry. Workers derive utility from consumption, and in one of the specifications developed below, also suffer disutility from working. Each firm has a single job that can be either filled or vacant and searching. Similarly, workers can be either employed by a firm or unemployed and searching. No new offers arrive while an agent is in a relationship. I abstract from capital accumulation and assume an exogenous rental rate of capital, c. The aggregate stock of capital, K, will be determined by demand.[4]

Assume meeting frictions can be represented by a function $m(u, v)$ that determines the instantaneous number of meetings as a function of the numbers of searchers on each side of the market, namely, unemployed workers u and vacancies v. Suppose m exhibits constant returns to scale and is increasing in both arguments. Let $q(\theta)$ denote the (Poisson) rate at which a vacancy contacts an unemployed worker, with $\theta = v/u$.[5]

Each firm has access to a technology $f(x, n, k)$ that combines the hours supplied by the worker it employs, n, and capital, k, to produce a homogeneous consumption good. The match-specific productivity level is stochastic and indexed by the random variable x. I assume that

$$f(x, n, k) = x \min(n, k) \tag{1}$$

and interpret k as the firm's capacity or scale of operation. Thus, output is linear in hours but is bounded above by the stock of capital the firm is operating with. The convention is that the technology is such that all projects have the same scale of operation k. Every firm has to rent and put in place k units of capital to be able to engage in search while vacant and to produce while filled.[6] This captures the notion that hours are a fully flexible factor, but capital is relatively fixed. Firms rent capital from a competitive market at flow cost c.

The process that changes the match-specific productivity, x, is Poisson with finite arrival rate λ. When a match of productivity x suffers a change, the new value x' is a draw from the fixed distribution $G(\cdot)$. So the productivity process is persistent (since $\lambda < \infty$), but—conditional on change—it is independent of the firm's previous state.[7] The Poisson process and the productivity draws are independently and identically distributed across firms, and there is no aggregate uncertainty. The focus will be on steady-state outcomes.

In the next section, I will show that there is a unique productivity level R_t such that active matches dissolve if productivity ever falls below that level and new matches form only if their initial productivity is at least R_t.[8] Let $H_t(x)$ denote the cross-sectional productivity distribution of active matches. That is, $H_t(x)$ is the fraction of matches producing at productivities x or lower

empirical measures of TFP for the ways of measuring aggregate inputs that can be found in the literature, and finally, this framework has already been used extensively to analyse the effects of similar policies on many other aggregate labour-market outcomes (see Ljungqvist and Sargent, 2000; Pissarides, 2000; and references therein).

4. This is the usual small, open-economy assumption. The model abstracts from saving and accumulation because the focus here is on isolating the effects of labour-market policies on the level of TFP. But even in the context of trying to explain income differences, Prescott (1998) and Parente and Prescott (2000) conclude that one cannot rely on policies that cause differences in saving rates, as they do not vary systematically with countries' incomes.

5. Note that $q(\theta) = m(1/\theta, 1)$, and hence $q' < 0$. The rate at which a worker contacts a vacancy is $\theta q(\theta)$ and is increasing in θ. For some of the existence proofs, assuming $\lim_{\theta \to 0} q(\theta) = \infty$ will also be useful.

6. The idea is that in order to search, the firm must have borrowed some capital, say, to set up a plant, and plants come in a single size, k.

7. This is the process used by Mortensen and Pissarides (1994). For reasons that will become clear shortly, Section 8 generalizes the model by specifying that when a match of productivity x suffers a change, the new value x' is a draw from the fixed distribution $G(\cdot \mid x)$. If $G(\cdot \mid x_1) < G(\cdot \mid x_0)$ when $x_0 < x_1$, then apart from being persistent, the idiosyncratic shocks are also positively correlated over time.

8. Mortensen and Pissarides (1994) work with a bounded support and assume new matches start off with the highest productivity. I relax these assumptions and treat active and new matches symmetrically. In the model considered here, the initial productivity of a match is a nondegenerate random variable drawn from the same distribution as the innovations to active matches.

© 2006 Federal Reserve Bank of Minneapolis

This content downloaded from
158.143.193.212 on Fri, 19 Nov 2021 13:09:47 UTC
All use subject to https://about.jstor.org/terms

at time t. The time path of $(1-u_t)H_t(x)$, the mass of matches producing at productivities x or lower at time t, is given by[9]

$$\begin{aligned}\frac{d}{dt}[(1-u_t)H_t(x)] = {} & \lambda(1-u_t)[1-H_t(x)][G(x)-G(R_t)]+\theta q(\theta)u_t[G(x)-G(R_t)] \\ & -\lambda(1-u_t)H_t(x)G(R_t)-\lambda(1-u_t)H_t(x)[1-G(x)] \\ & -\delta(1-u_t)H_t(x).\end{aligned}$$

The first term accounts for the matches with productivities above x that get innovations below x but above R_t. The newly formed matches that start off with productivities no larger than x are in the second term. The third term represents the matches with productivity in the interval $[R_t, x]$ that get shocks below R_t and are destroyed. The fourth term accounts for those matches in the same interval that "move up" by virtue of having drawn productivities larger than x. The parameter $\delta < \infty$ denotes the arrival rate of an independent Poisson process that causes separations for reasons unrelated to the match-specific productivity shocks. The last term accounts for the matches in the interval $[R_t, x]$ that are destroyed for exogenous reasons.[10] In steady state,

$$\begin{aligned}H(x) &= \left[\frac{\lambda}{\delta+\lambda}+\frac{\theta q(\theta)u}{(\delta+\lambda)(1-u)}\right][G(x)-G(R)] \\ u &= \frac{\delta+\lambda G(R)}{\delta+\lambda G(R)+\theta q(\theta)[1-G(R)]}. \qquad (2)\end{aligned}$$

Using (2), the steady-state cross-sectional productivity distribution becomes

$$H(x) = \frac{G(x)-G(R)}{1-G(R)}. \qquad (3)$$

Next, I consider the problems faced by a worker and a firm. The values of unemployment and employment to a worker are denoted U and $W(x)$, respectively, and solve

$$rU = b+\theta q(\theta)\int \max[W(z)-U,0]dG(z) \qquad (4)$$

$$rW(x) = w(x)+\lambda\int \max[W(z)-U,0]dG(z)-(\delta+\lambda)[W(x)-U], \qquad (5)$$

where r is the discount rate, $b \geq 0$ denotes a worker's flow income while unemployed, and $w(x)$ is the wage earned by a worker employed in a match of productivity x.[11]

Firms can be either filled or vacant and searching. The problem of a searching firm is summarized by

$$rV = -ck+q(\theta)\int \max[J(z)-V,0]dG(z), \qquad (6)$$

where V is the asset value of a vacancy and $J(x)$ the asset value of a filled job with productivity x. Letting $\pi(x)$ denote flow profit, $J(x)$ satisfies

$$rJ(x) = \pi(x)+\lambda\int \max[J(z)-V,0]dG(z)-(\delta+\lambda)[J(x)-V], \qquad (7)$$

9. The fact that active matches will form and continue only for productivities at least as large as R_t means that $H_t(R_t) = 0$. Thus, the derivation focuses on $x \geq R_t$.

10. In Section 8 I extend the analysis to the case in which the rate δ is a decreasing function of the idiosyncratic productivity parameter x. I will discuss the reasons why this extension may be worth exploring shortly.

11. This formulation assumes the worker suffers no disutility from work; but see Appendix B.

© 2006 Federal Reserve Bank of Minneapolis

This content downloaded from
158.143.193.212 on Fri, 19 Nov 2021 13:09:47 UTC
All use subject to https://about.jstor.org/terms

where $\pi(x) = x\min(n,k) - w(x) - ck - \phi n - C(x,\phi)k$. Instantaneous profit is the residual output after the wage $w(x)$, and all other costs of production have been paid out. This formulation has three such costs: the rental on capital, ck; a variable cost, ϕn, that can be managed by varying hours; and a fixed cost C per unit of capital.[12] The variable cost ϕn and the fixed cost Ck are introduced to allow for the possibility of labour hoarding and under-utilization of capital, two pervasive features of the data. In the presence of these costs, for some parametrizations it will be possible that at low productivity realizations, the firm may choose to keep the worker employed despite requiring that she supply zero hours. In Sections 4 and 6.2 I will show that this type of labour hoarding has interesting aggregate implications when it occurs in equilibrium.

To fix ideas, one can think of ϕ as the cost of electricity, for instance, with electricity usage being proportional to hours worked. Alternatively, in Appendix B I show how to modify (5) and $\pi(x)$ to get an equivalent formulation in which ϕn is the worker's disutility from supplying n hours to her employer. Either way, the key observation is that $\phi > 0$ is necessary for the model to display the type of labour hoarding described above. Intuitively, if the marginal product of labour, x, is lower than its marginal cost, ϕ (expressed either in terms of resources to the firm under the first interpretation or in terms of disutility to the worker under the second), then efficiency will require the match to set $n = 0$: the worker should not supply hours to the firm. Under these circumstances, whether the firm and the worker should choose to preserve the match is a different matter. One possibility is that the instances in which the firm–worker pair sets $n = 0$ are also the cases in which the pair chooses to dissolve the match. Alternatively, one could imagine that the pair may choose $n = 0$, yet decide to preserve the match, anticipating that the low realization of the idiosyncratic shock may be reversed soon. In the first scenario no hoarding is exhibited. In the latter it is. The fixed cost C introduced in this section makes it possible for the second scenario to arise in an equilibrium with endogenous separations.

Notice that if $C = \phi = 0$, then the model reduces to the standard set-up of Mortensen and Pissarides (1994). For this special case, all the results on how labour-market policies affect the level of TFP still apply even though the equilibrium does not exhibit hoarding. In this sense, neither of these costs is essential. However, in the presence of the variable cost ϕ and the fixed cost C, the model is also able to deliver equilibrium hoarding, and as it turns out, this feature has interesting implications for how the aggregate production function looks.

Since the fixed cost C is perhaps the only non-standard element of the model, several remarks are in order. First, at a technical level, incorporating a fixed-cost specification that is decreasing in the shock x is introduced in this section as a simple device to avoid a "flat spot" in flow profit that would otherwise carry over to the value functions.[13] Second, in Section 8 I drop the fixed cost and redo the analysis in a version of the model in which the draws of the idiosyncratic shock are correlated over time. In another extension explored in Section 8, I again drop the fixed cost and show that the key insights also apply in a version of the model in which the destruction rate δ is decreasing in the productivity shock. I wish to emphasize that these alternative specifications seem natural (perhaps even more so than the usual benchmark in which the shocks hitting the match are independent of its state) and are able to deliver hoarding without the fixed cost C.[14] Given all this, and for expositional purposes, I will—for the time being—use a particularly convenient specification for the fixed cost, namely, $C(x,\phi) = \max(\phi - x, 0)$.[15]

12. The cost C is fixed in the sense that the firm can avoid it only by shutting down, but—as the notation indicates—it may depend on the realization of the idiosyncratic shock and possibly also on the parameter ϕ. More on this is given below.

13. See Appendix B, in particular, the discussion around equation (47), for details.

14. Flow profit has a flat spot in both of these alternative formulations, but the serially correlated shocks ensure that the value functions do not inherit this flat spot.

15. This formulation is convenient because it will imply that the flow profit is affine and strictly increasing in x. Since—even accepting the presence of a fixed cost—this particular formulation may seem somewhat contrived, in

© 2006 Federal Reserve Bank of Minneapolis

This content downloaded from
158.143.193.212 on Fri, 19 Nov 2021 13:09:47 UTC
All use subject to https://about.jstor.org/terms